CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · DPP SHEET

Chapter 16: Loci (Locus and its Constructions)

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. State the locus of a point that moves so that it is always 5 cm from a fixed point O.

2. State the locus of a point equidistant from two fixed points P and Q.

3. State the locus of a point equidistant from two parallel lines 8 cm apart.

4. State the locus of the centres of all circles that pass through two fixed points A and B.

5. State the locus of a point inside $\angle ABC$ that is equidistant from BA and BC.

6. State the locus of the centre of a coin of radius 1 cm that rolls along the inside edge of a circular tray of radius 10 cm.

7. Q lies on the bisector of $\angle ABC$. The perpendicular distance from Q to AB is 3.2 cm. Find its perpendicular distance from BC.

8. Name the point that is equidistant from the three sides of a triangle, and say how it is constructed.

9. Write the steps to locate a point equidistant from the three vertices of $\triangle ABC$.

10. $AB = 7$ cm. Describe how to find the points that are 5 cm from B and 2 cm from the line AB.

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Answer Key

1. The circle with centre O and radius 5 cm
2. The perpendicular bisector of PQ
3. The line parallel to both, 4 cm from each
4. The perpendicular bisector of AB
5. The bisector of $\angle ABC$
6. A circle of radius 9 cm concentric with the tray
7. 3.2 cm
8. The incentre; the meeting point of the bisectors of any two angles
9. Draw the perpendicular bisectors of any two sides; their meeting point is the circumcentre.
10. Draw the circle with centre B and radius 5 cm, and the two lines parallel to AB at 2 cm; they meet in four points.

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